



Graph-based effective knowledge tracing via subject knowledge mapping

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Abstract

Intelligent technology plays a pivotal role in revolutionizing learning assessments, overcoming the constraints of traditional assessment methods and driving educational innovation. Knowledge tracing (KT) emerges as a critical component for assessing students' learning states and forecasting their future performance. However, existing graph-based KT models often ignore certain real-world factors (e.g., sequence dependencies, question information, and hierarchical relationships between knowledge concepts). These factors hinder the model's overall performance in predicting students' learning progress and identifying their knowledge deficiencies, thus limiting the model's applicability in personalized learning. To address these challenges, we propose a novel graph-based effective knowledge tracing model via subject knowledge mapping (SGKT). We begin by extracting essential features from historical learning interactions, which include the relationships between questions, the difficulty of questions, and the relationships between knowledge concepts. These knowledge concepts are seamlessly integrated with a purpose-designed subject knowledge mapping, establishing intricate hierarchical relations. Subsequently, we employ graph convolutional networks to aggregate the embeddings of questions and knowledge concepts, thereby optimizing the update process for predicting students' knowledge states. Experimental validation on the dataset demonstrates the superior performance of our model, surpassing the original KT model and other relevant approaches by up to 5% in terms of the area under the curve and accuracy. These results not only demonstrate the efficacy of our proposed model but also provide valuable insights for advancing learning assessments.

Keywords Learning assessment · Knowledge tracing · Graph convolutional network · Subject knowledge mapping · Feature embedding

1 Introduction

UNESCO, in *Education 2030 Framework for Action*, underscores education's commitment to individuals' holistic development (Mundial, 2016). Personalized education hinges on students understanding their knowledge mastery levels (Williams & Zipser, 1989; Hooshyar, 2024). As Ausubel et al. (1978) noted, "the single most important factor shaping the learning process is what the student is already aware of". The ascent of personalized learning is propelled by technological advancements in Artificial Intelligence (AI), big data, and the Internet of Things (IoT). In this technological environment, Knowledge tracing (KT) techniques have emerged in the realm of human–computer interaction, designed to simulate the dynamic interactions between students and the exercise content they engage with. This simulation enables accurate, real-time assessment and prediction of students' changing knowledge mastery levels (Abdelrahman et al., 2023). Consequently, students can address weaknesses, identify areas for improvement, and choose suitable methods and resources for learning. Simultaneously, teachers can dynamically tailor teaching activities based on the insights provided by KT, ensuring that classroom designs are in harmony with the actual learning conditions of students, thereby fulfilling the goals of teaching support.

The growth of personalized learning underscores the imperative to identify and dynamically assess students' knowledge states (Murtaza et al., 2022). Given that students' knowledge states are often implicit, direct assessment poses challenges (Ma et al., 2023). Figure 1 depicts the shift from "one-dimensional static outcome-based diagnosis" to "multi-dimensional dynamic process-based Tracing". Initially, Classical Test Theory (CTT) assessed students' ability ratings directly through fixed exams and their corresponding scores (Gulliksen, 2013). However, the approach of directly linking constant difficulty levels of each exam with ability scores lacks scientific rigor. To address this issue, Item Response Theory (IRT) emerged, facilitating the estimation of mastery stages of latent-knowledge (Maghsudi et al., 2021; Traub, 1981). IRT considers students' abilities and test paper conditions, creating a functional relation. However, relying solely on a standard total score still fails to characterize students' mastery of various knowledge concepts due to individual cognitive variances. This implies that even if the total scores are the same, there can be significant differences in students' understanding of specific knowledge concepts.

In the realm of assessment theory, a new paradigm emerged with the creation of cognitive diagnostic theory, integrating psychological assessment and cognitive science (Ackerman, 2014; Lubis & Yus, 2024). Gierl et al. (2008) utilized cognitive attribute diagnostics to uncover variations in students' arithmetic scores, offering comprehensive assessments of algebraic understanding. Static cognitive diagnosis suits situations like exams, providing in-depth assessments of learning data within a specific time-frame, indicating the current states of knowledge mastery. However, real learning unfolds gradually over an extended period. Through a continual process of learning, forgetting, and responsive feedback, students' command of knowledge undergoes dynamic changes (Liu et al.,

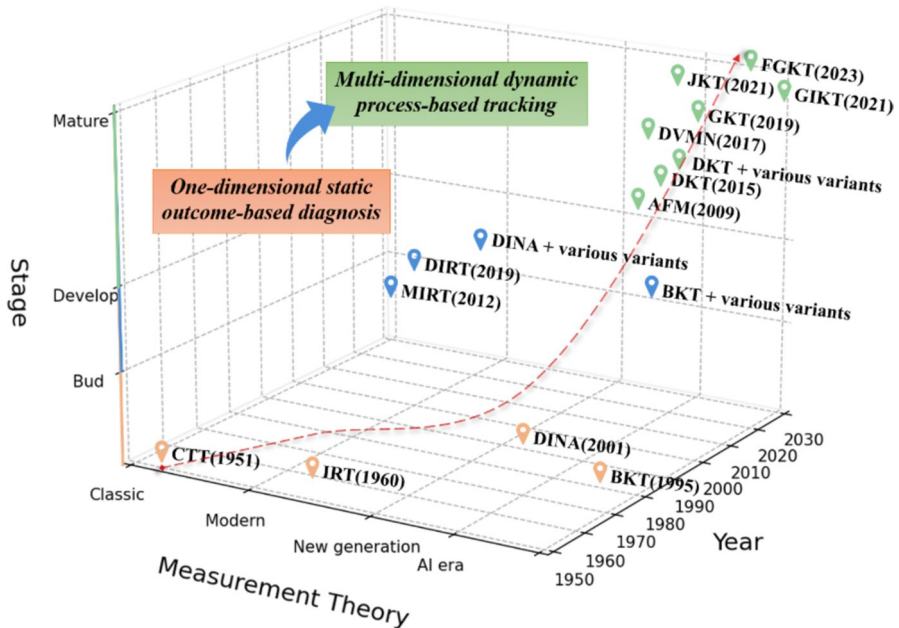


Fig. 1 The evolution of learning assessment theory. Each theoretical model goes through three stages of evolution, which may be seen by looking at learning assessment theory: bud, develop, and mature

2019). Consequently, dynamic cognitive diagnostics, also known as knowledge tracing (KT), have been proposed. Suited for tracing changes in knowledge states, KT utilizes knowledge acquisition methods. Figure 2 illustrates the process of knowledge tracking based on historical learning interactions. KT tasks are used to dynamically, and accurately diagnose students' knowledge mastery and forecast how well students will perform on the task to come.

Based on different machine learning approaches, scholars categorize existing KT research into three types: Bayesian Knowledge Tracing (BKT), Additive Factor Model (AFM), and Deep Knowledge Tracing (DKT). DKT is a promising approach that can significantly improve predictive performance compared

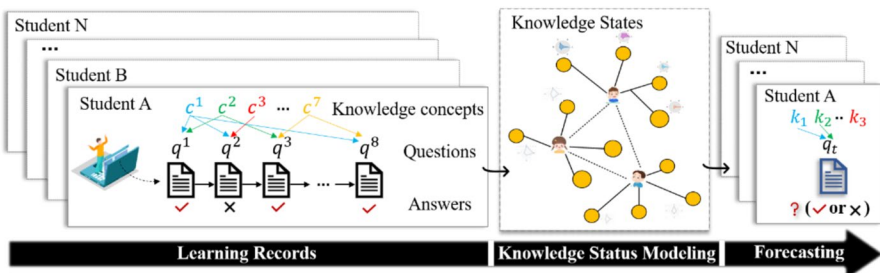


Fig. 2 A simple example of knowledge tracing process

to traditional models. However, DKT has some inherent issues, such as neglecting long-term dependencies in learning sequences (Yeung et al., 2018) and not providing a comprehensive modeling of complex interactions. To overcome these challenges, graph neural networks (GNN) have been introduced as a novel approach for KT. A graph-based knowledge tracing (GKT) was introduced by Nakagawa et al., which improves performance and interpretability in capturing the connections between students' knowledge concepts and cognitive state models. (Nakagawa et al., 2019; Battaglia et al., 2018). Building on this, exploring important information such as students' behaviors, knowledge concepts, and question sequences has become feasible. Examples include the joint graph convolutional network based deep knowledge tracing (JKT) (Song et al., 2021), graph-based interaction model for knowledge tracing (GIKT) (Yang et al., 2020), and graph attention neural network model with behavior features for knowledge tracing (GAKT-BF) (Zhang et al., 2023).

However, the existing models have certain limitations. On the one hand, these models do not take into account the relationship between questions and knowledge concepts, nor the effect that questions' level of difficulty has on students' learning (Qiu et al., 2024). On the other hand, current methods often reduce practice records to a simple sequence of knowledge concepts, failing to fully explore the various potential relations between knowledge concepts, thus necessitating a more rational knowledge concept representation. To fill the gap, we propose a graph-based effective knowledge tracing model via subject knowledge mapping (SGKT). Firstly, we finely partition various knowledge concepts in the subject, identifying possible structural connections between isolated knowledge concepts by considering students' historic learning interactions and subject knowledge characteristics. Subsequently, we hierarchically classify the concept mapping question, constructing a comprehensive subject knowledge mapping (SKM) as a robust foundation for subsequent KT. By integrating information from both long-term dependency relations and neighboring nodes, we employ graph convolutional networks (GCN) to aggregate question and knowledge concept embedding, transforming the KT task into a graph-based representation. As students' knowledge states evolve, we learn relations between fundamental concepts from the dynamic latent concept graph and update the knowledge states graph. This enables us to accurately capture students' mastery of various knowledge concepts, significantly enhancing the performance and interpretability of KT.

The main contributions of this paper can be summarized as follows:

1. We propose a novel conceptual framework for subject knowledge mapping, along with its automated generation process. By employing "*Big Ideas Math*" as an example, our model adopts a granular, hierarchical approach to organizing knowledge concepts and questions. This method not only refines the representation of knowledge concepts but also enhances the ability to forecast students' subject knowledge states through the integration of an intricate information architecture. Our model can overcome the representational shortcomings inherent in conventional KT models, providing fresh insights and methodologies for accurately assessing students' mastery of subject matters.

2. We propose a graph-based effective knowledge tracing model via subject knowledge mapping (SGKT). Our innovative approach goes beyond simply integrating subject knowledge with the characteristics of KT models. Specifically, it leverages GCN to learn node embeddings within the graph and the handling of information transmission between adjacent nodes. We can accurately represent students' knowledge states in the form of a directed graphs. This novel method not only enhances the precision of dynamic tracing and prediction of students' learning progress but also strengthens the model's ability to capture and understand complex interactions within students' knowledge networks.
3. We thoroughly evaluated the performance of our SGKT model in comparison with the prevailing deep knowledge tracing and graph-based methods, including DKT, DKVMN, and GKT, among others. The SGKT model surpassed these established methods across essential performance metrics, notably in enhancing prediction accuracy, accurately monitoring the dynamic shifts in students' knowledge states, and elucidating the interactions among subject knowledge concepts. These results underscore not only the SGKT model's superior capacity for deep comprehension of students' knowledge states but also its innovation and efficacy in precisely navigating the learning trajectory.

The rest of this paper is organized as follows. The related work is discussed in Section 2. Section 2 discusses the related work. Section 3 defines the KT problem, constructs SKM and explains details in our model. Section 4 discusses the related dataset, baselines, experimental design, results, and application prospects. Finally, Section 5 discusses the conclusions.

2 Related work

In this section, we first conduct a survey of various KT models, delving into the details of graph-based KT. Then, we introduce the subject knowledge mapping.

2.1 Knowledge tracing

KT models have garnered significant interest from scholars across computer science, education, and psychometrics. Each model type exhibits distinct advantages, disadvantages, performance characteristics, and application scopes (Abdelrahman et al., 2023; Song et al., 2022), summarized in Table 1.

As early as the 1990s, Corbett and Anderson (1994) presented the BKT model, which was based on the hidden markov model (HMM). By observing students' responses (correct or wrong), this method simulates the complete process of acquiring knowledge concepts, presuming constant abilities of students. Nonetheless, it might not be appropriate to assess students' knowledge states using discrete probabilities because knowledge concepts are not totally independent. A number of logistic regression models were then put forth. Assuming variations in the abilities of students, Cen et al. (2006) proposed AFM based on the logistic regression (LR). The

Table 1 Comparison of main features of KT models

Model	Theory	Hypothesis	Pros	Cons
BKT	HMM	Discrete transition	Simple model, Interpretability	Single knowledge concept, Knowledge states tuple representation, Failure to consider students' knowledge forgetting, Ignore the relation between knowledge concepts, Students share a set of parameters regarding knowledge concepts, No limit on parameter size, Ambiguous mapping between hidden states and exercises
AFM	LR	Gradual shift	Simple model, Add Q matrix, Interpretability, Multiple knowledge concepts	Rely on expert annotation, Failure to consider that the states of knowledge changes dynamically, Difficult to obtain students' knowledge states in an extended period of time, Sensitive to multicollinearity of independent variables
DKT	RNN	None	Better performance, Solve expert dependencies, Multiple knowledge concepts	Cold start problem, Large amount of training data, Complex model, lack of explainability, Failure to consider learning features, Failure to explore exercises' characteristics, Unable to capture long sequence of dependencies, Ignore the logical relation between knowledge concepts, Students' mastery of knowledge concepts is discontinuous
GKT	GNN	None	Better performance, Dynamic adaptability, Model complex relations, Multiple knowledge concepts	Failure to consider student characteristic information, Failure to explore exercises' characteristics, Ignore the logical relations between knowledge structures, Ignore the dependencies between knowledge concepts and exercises

logistic function represented by these parameters, which are specified for distinct ideas that students know, directly predicts the likelihood that students will provide correct responses.

However, this method is dependent on theoretical presumptions and necessitates classification of domain knowledge by professionals, which may be biased and constrained. DKT model based on RNN (Williams & Zipser, 1989) presented by Piech et al. (2015), addresses some of these issues in the context of the deep integration of artificial intelligence and education. From historical learning interactions, RNN can directly train continuous and hidden features that indicate students' knowledge state, improving performance on prediction. To address the issue of DKT's inability to accurately determine students' mastery of knowledge concepts, the dynamic key-value memory network (DKVMN) borrows the core idea from DKT (Zhang et al., 2017). However, it improves upon it by employing two matrices to separately preserve and adjust the mastery levels of the associated knowledge concepts. From the standpoint of data structures, KT tasks frequently exhibit a variety of relation structures, including those based on knowledge concept similarity, knowledge concept dependence, question-knowledge correspondence, etc. Therefore, GKT based on GNN was proposed by Gori et al. (2005). After being reformulated as a time series node-level classification challenge, the KT task was resolved by the use of traditional graph learning techniques like GNN information transferring.

Existing research graph-based KT invariably reveals some drawbacks and restrictions (Zanellati et al., 2024). It does not take into account question characteristics, higher-level dependence information between knowledge concepts and questions, relations between knowledge concepts, and student characteristic (Anzanello & Fogliatto, 2011). Concurrently, it becomes difficult to track long-term correlations between sequences when using RNN to model student learning interactions. Problems like gradient disappearance or explosion can occur in prolonged learning settings, which can result in worse prediction accuracy and higher computing complexity (Zhu et al., 2020).

2.2 Subject knowledge mapping

As mentioned in the *Knowledge Maps Development Report (2022)*, we are at a critical crossroads where our cognitive intelligence is replacing our perceptual intelligence. Knowledge mappings play a vital role in cognitive intelligence, characterized by revealing relations between pieces of knowledge concepts. They depict students' trajectories on specific knowledge mappings, providing support for personalized and intelligent learning. Divergent concepts such as SKM (Hanewald & Ifenthaler, 2014), cognitive mapping (Jameson, 1988), and curriculum mapping (Jacobs, 2004) have emerged due to differences in their purposes, connotations, and applications. Odonnell et al. (2002) defined it as a node-link representation, where entities are connected through labeled links. Lei et al. (2006) considered SKM as semantic networks connecting knowledge concepts and resources. Li et al. (2019) viewed it as a presentation of knowledge content in the

form of courses. Wu and Wang (2023) described SKM as structures that integrate resources through a mesh, showcasing connections between knowledge concepts.

The theoretical underpinning traces back to Kant's schema theory, asserting that schemata serve as the link between concepts and perceptual objects, and new concepts gain meaning through relations with existing knowledge (Melnick, 2001). Consequently, we posit that the objective of SKM should center around cultivating student competencies, grounded in revealing the relations within subject knowledge. By utilizing the question-solving process as an intermediary, concrete mappings are established between knowledge concept systems and corresponding question-solving abilities. This results in a systematic, hierarchical framework encompassing knowledge concepts, questions, and question-solving abilities, furnishing a lucid visual representation tool that accentuates the inter-connected relations among them.

The definition of component elements also varies slightly based on the various training objectives of students, including knowledge acquisition, resource suggestion, and ability development. We propose that the Fig. 3 represents the structural model. Based on the knowledge classification proposed by modern cognitive psychologist Anderson and Krathwohl (2001) and the refined classification by American scholar Woolley et al. (2019), knowledge concept structures are categorized into declarative knowledge, procedural knowledge, and conditional knowledge. Declarative knowledge encompasses facts and concepts, while procedural knowledge comprises programs, rules, and steps. Conditional knowledge includes strategies and innovation. Correspondingly, the associated question structures are stratified into rule-based, application, and probing questions, ranging from easy to difficult. Furthermore, the question-solving ability structures align with Bloom's taxonomy of cognitive objectives (Hubalovsky et al., 2019), encompassing six sub-abilities. This offers a thorough framework for comprehending and evaluating the several knowledge dimensions and question-solving abilities inside SKM.

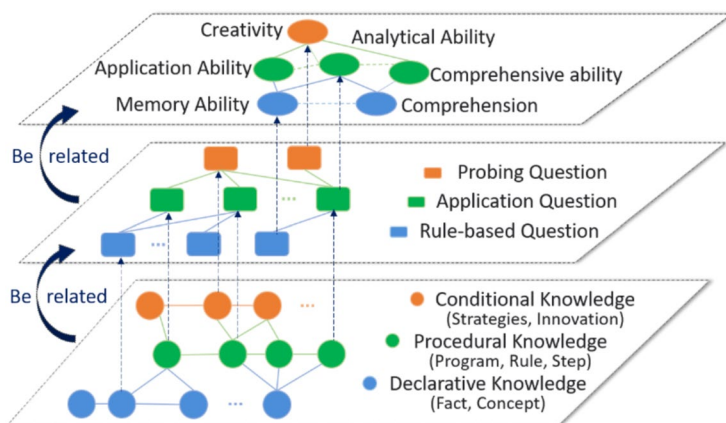


Fig. 3 The structure model of subject knowledge mapping. It divides into three categories: knowledge concepts, associated questions, and the question-solving abilities

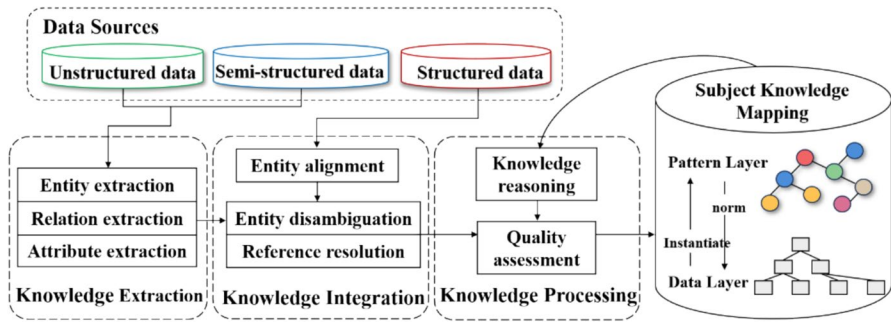


Fig. 4 The construction process of SKM. Three main categories of sources are used for data gathering. On these three categories of data, intelligent technology is used to perform basic operations including knowledge extraction, integration, and processing (Ke & Lin, 2022)

According to the previously described research, SKM has undergone developmental stages that include manual, crowd-sourced, and automatic building, all with the aim of obtaining subject cognitive intelligence. In Fig. 4, the top-down construction strategy is demonstrated. On top of this base, topic question and the systems that support question-solving abilities are created, and pertinent knowledge concepts are linked to create an all-encompassing SKM.

3 Model structure

This section will contain details about the SGKT model. Firstly, SGKT model is briefly introduced and the relevant definitions of KT are explained. We then take over how to build a SKM. The knowledge domain characteristics and the knowledge tracing model can then be effectively integrated once the knowledge concepts and question characteristics have been embedded and represented. The represented data, along with historic learning interactions, is sent to KT model in combination. Finally, a graph-based KT model that integrating SKM is proposed. This model incorporates relations among knowledge concepts and features of questions into KT, further enhancing interpretability and predictive performance of the results.

3.1 Model overview

With the widespread application of digital technologies represented by artificial intelligence, knowledge mappings, and cloud-integrated smart learning systems, practical support conditions and implementations have been provided for dynamically assessing students' learning states and existing problems. Therefore, we construct a SGKT model and Fig. 5 presents the architecture. The procedures are described in full in the paragraphs that follow.

Our model is composed of four parts. Using the American middle and high school mathematics course "*Big Ideas Math*" as an example, we first establish a SKM. Subject's knowledge concepts are arranged hierarchically based on the knowledge

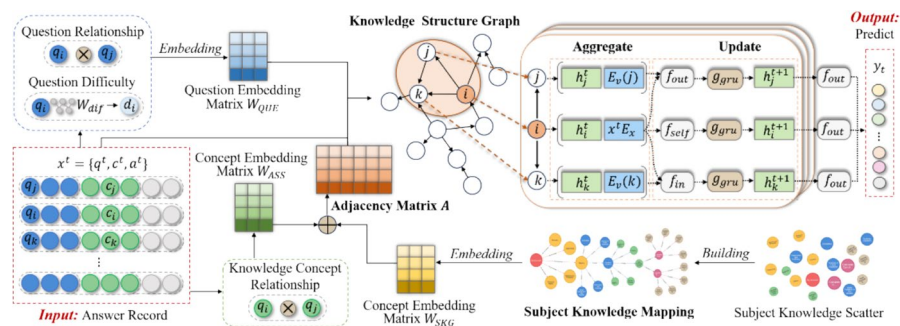


Fig. 5 The overall framework of graph-based effective knowledge tracing via subject knowledge mapping

concept structure shown in Fig. 3. Then combine previous interaction information with the created SKM. To store, handle, and display graph data as directed graphs, use the Neo4j graph database. Second, to improve question representation, include question relations and difficulty. More precisely, we propose a loss function to enhance the question representation and treat the relation and difficulty of questions as two training objectives. The details about the question and knowledge concepts are subsequently integrated into the next KT model.

Then, to solve the vanishing gradient problem in traditional RNN, a recurrent layer GRU is used to generate the hidden states of knowledge nodes to better capture long-term dependencies in sequence data. And GCN is used to aggregate information using the hidden states of the current node and its neighbor nodes at each layer. Then, when the knowledge concept transmits the message, GCN is used to update the hidden state of the knowledge concept. Among them, the contribution of each neighbor node is weighted and combined into the representation of the current node, which can more comprehensively understand the relation between nodes in SKM and help solve complex pattern recognition and information transfer problems in KT tasks.

Finally, predict students' mastery of each knowledge concept. These prediction results can not only be used to assess students' overall learning level, but also provide personalized feedback for each knowledge concept to help students improve the corresponding subject question-solving abilities in a more targeted manner.

3.2 Problem definition

KT aims to track students' cognitive states in real time based on their historical learning interactions and predict their future performance in answering questions. It can be formulated as $y^t = f(x^1, \dots, x^t)$, where $x^t = \{q_i^t, c_j^t, a^t\}$ is a tuple that considers an answered exercise q_i^t , c_j^t denotes the knowledge concept j that the exercise i covers, and whether the exercise i was answered correctly $a^t \in \{0, 1\}$ at time step t , y^t is the probability of the student answering each exercise correctly at the next time step $t + 1$, and $f()$ is the KT model. GKT recasts the historical learning interactions

as a graph structure $G = (V, E)$, involving N knowledge concepts, called nodes N , known as nodes $V = \{v_1, \dots, v_N\}$, and these knowledge concepts share dependency relations, known as edges $E \subseteq V \times V$.

And it applies GCN to handle the KT task as a time series node prediction problem. At time step t , we assume that a student has separate temporal knowledge states for each knowledge concept, $h^t = h_{i \in V}^t$. Over time, this state of knowledge is updated. This knowledge states are updated over time. when the student solves an exercise associated with knowledge concept v_i , then the student's knowledge states for the answered concept itself h_i^t and for its related concepts $h_{j \in N_i}^t$ is updated. Here, N_i denotes a set of nodes neighboring v_i . Then, we effectively integrate the embedded representation of SKM and learning interactions, and jointly feed them into the graph-based KT model. In the discussion that follows, we will present our model for forecasting students' performance in the future.

3.3 Construction of subject knowledge mapping

The dataset for this research originates from the foreign online education platform Assistments, comprising processed data related to middle and high school mathematics. It is utilized to support specific educational research projects. The foundational work for constructing SKM relies on publicly available standard documents from foreign educational practices. The data is obtained through manual extraction, selecting authoritative academic course materials such as the "*Big Ideas Math*" series.

The data obtained are mainly unstructured and semi-structured data. We use the global vectors (GloVe) (Eke et al., 2021) for word representation model to process unstructured data. Through word segmentation, part-of-speech tagging, word vector initialization and training (Liu et al., 2018; Shi et al., 2018), it aims to identify entities and optimize the processing of information. For semi-structured data, we use the latest automatic feature engineering technology to eliminate noise and redundant information in the data through feature selection and generation methods (Alam et al., 2022; Wu & Wang, 2023). Use adaptive feature learning algorithms for data cleaning and annotation, and optimize the feature space through automated feature evaluation. SKM should not only present the knowledge itself, but also show the correlation between knowledge and form a knowledge network, which is highly consistent with the learning theory of connectivism (Zhu et al., 2016).

Based on the definition of various relations in topic features, manual annotation is employed. In order to construct a SKM, based on knowledge structure theory and subject characteristics, we divide the knowledge organization level into three levels: chapters, sections, and knowledge concepts (Liu et al., 2021). Through three rounds of annotation work, the hierarchical structure is first determined, and then the association between knowledge is defined through three relations. In the three rounds of construction, if the two annotations of the same set of knowledge relations are consistent, they are retained, otherwise they are eliminated to form the second round of construction results. The third round of review is conducted to ensure the accuracy and usability of the annotations. Finally, the

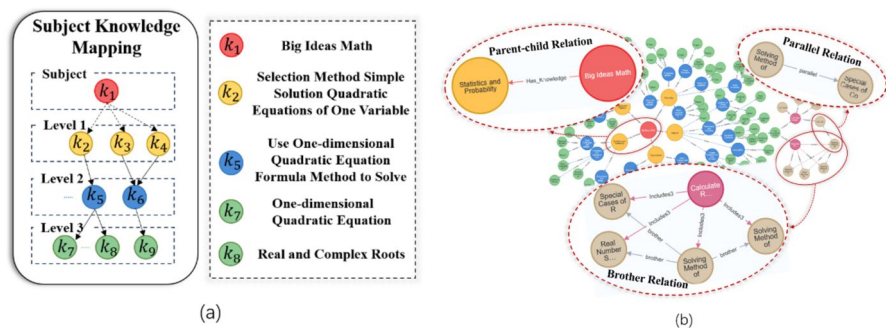


Fig. 6 **a** Represents creating knowledge concept structure. Consider the k_2 (Selection method simple solution quadratic equations of one variable) concept at the first level as an illustration. The level that follows k_2 (Use one-dimensional quadratic equation formula method to solve) at the second level is k_2 . k_3 develops two new concepts (One-dimensional quadratic equation, real and complex roots) at level 3. **b** Visualization of SKM. The relations between knowledge concepts are defined as three types of relations: parent–child, brother, and parallel

knowledge states under the parent–child relation, brother relation and parallel relation was obtained, as shown in Fig. 6 (a), and the construction of SKM was completed. Research focuses on adopting mature technologies rather than innovating on technical details. With the help of the Neo4j graph database, we store and process graph data and visualize it in the form of a directed graph, as shown in Fig. 6 (b). At the same time, it is represented by a two-dimensional matrix $W_{SKM} \in R^{N_1 \times N_1}$.

3.4 SKM embedding

Based on the SKM formed in the previous section, we considered integrating the knowledge concept relations in the SKM into knowledge tracing approach. The students' historical learning interactions and relations between knowledge concepts are fused, and the fused common representation is used as the input for KT. We select the ASSISTments2009 dataset as students' learning interactions. It includes main fields such as question number, student number, knowledge concept number, the number of times students requested prompts, the length of time students studied knowledge concepts, and students' answer results. When processing the data, we extract all "skill_name" and calculate the correlation weight between knowledge concepts through the cosine similarity measure function. Only knowledge concepts with similarity greater than 0.5 are selected to establish connections to form graph-structured data. Through multiple rounds of experimental verification, we found that when the similarity threshold is set to 0.5, the density of the graph structure and the performance of the model can be effectively balanced. And represented by adjacency matrix $W_{ASS} \in R^{N_2 \times N_2}$. The specific formula of the similarity measure function can be formulated as:

$$W_{ASS(i,j)} = \begin{cases} \frac{v_i \cdot v_j}{\|v_i\| \cdot \|v_j\|} & i \neq j \\ 0, & i = j \end{cases} \quad (1)$$

Among them, $v_i \cdot v_j$ represents the inner product of vectors, and $\|v_i\|$ and $\|v_j\|$ represent the norms of vectors v_i and v_j respectively.

Then, we associate the knowledge graph in the students' learning interactions with SKM to form a new graph, which is represented by the adjacency matrix A :

$$A \in R^{(N,N)} = \begin{bmatrix} W_{SKM} & S_{SKM-ASS} \\ S_{SKM-ASS}^T & W_{Ass} \end{bmatrix} \quad (2)$$

N is equal to the sum of $N_1 + N_2$. The correlation weight between the knowledge concepts of SKM and ASSISTments dataset is calculated through the cosine similarity measure function, represented by $S_{SKM-ASS(i,j)} = \frac{v_i \cdot v_j}{\|v_i\| \cdot \|v_j\|}$.

In the subsequent graph convolution operations, matrix A is employed to compute the relation between node v_i and other nodes v_j for updating the representation of node v_i . Each node v_i undergoes updates relying on the features of its connected nodes and the weights from matrix A . This iterative process enhances the model's ability to capture intricate relations among nodes in GCN, thereby yielding more informative node representations.

3.5 Question embedding

Relying solely on knowledge concepts for predicting students' performance overlooks the inherent distinctions among exercises, leading to a diminished performance of KT model. Acknowledging that exercises encompass varying difficulty and semantic nuances, we employ question embedding to delve into effective representations of exercise's questions. To account for this, we incorporate the difficulty of a question as additional information. We utilize a fully connected layer to transform the representation vector of the question q_i into a scalar value $\widehat{qd}_i$, representing its difficulty, as follows:

$$\widehat{qd}_i = q_{qi} W_{dif} \quad (3)$$

Here, W_{dif} represents the weight matrix of the network layer. In general, a exercise's question with a high correct answer rate is considered easy, indicating that it is easily solvable. qd_i is the difficulty level label of question q_i , which is used as training corpus. The loss function Ld is employed to optimize the representation of the question following Eq. (4), ensuring that the difficulty labeling results assigned by subject experts and renowned teachers align with the training outcomes.

$$Ld(\widehat{qd}_i - qd_i) = \begin{cases} 0.5(\widehat{qd}_i - qd_i)^2, & \text{if } |\widehat{qd}_i - qd_i| < 1 \\ |\widehat{qd}_i - qd_i| - 0.5, & \text{otherwise} \end{cases} \quad (4)$$

When $\widehat{qd}_i - qd_i < 1$, where the error value is less than 1, it doesn't easily lead to gradient explosion. In such cases, it reverts to the mean square error loss form and is assigned a smoothing coefficient of 0.5. However, when $|\widehat{qd}_i - qd_i| \geq 1$, the potential for gradient explosion increases. In these instances, we decrease the power of the loss to prevent gradient explosion.

Each exercise's questions are annotated with relevant knowledge concepts, capturing relations based on these concepts. For instance, questions like " $x^2 + 2x + 2 = 9$ " and " $x^2 - 3x + 9 = 18$ " both pertain to the "*quadratic equations of one variable*" knowledge concept. Students who answer " $x^2 + 2x + 2 = 9$ " correctly are likely to answer " $x^2 - 3x + 9 = 18$ " correctly as well, indicating similarity. Questions may involve multiple knowledge concepts, and the degree of overlap contributes to varying degrees of similarity. We utilize the Jaccard similarity coefficient to calculate the similarity, denoted as qs_{ij} , between question q_i and question q_j , as follows:

$$qs_{ij} = \frac{|C_{q_i} \cap C_{q_j}|}{|C_{q_i} \cup C_{q_j}|} \quad (5)$$

Among these, C_{q_i} represents the number of knowledge concepts within question q_i . To enhance the capture of questions' correlations during training computations, we employ the Bert-Sim method. This involves calculating the cosine similarity of the BERT embedding for two questions and establishing connections between questions with higher similarity by setting a threshold (Zhou et al., 2022), thereby forming an association graph. The resultant graph proves useful in subsequent knowledge tracing tasks for a more nuanced understanding and utilization of relations between questions. Assuming their BERT embedding are denoted as E_i and E_j respectively, the cosine similarity $\widehat{qs}_{ij}$ can be calculated using the following:

$$\widehat{qs}_{ij} = \frac{E_i \cdot E_j}{\|E_i\| \cdot \|E_j\|} \quad (6)$$

Here, $\cdot$ represents the dot product of vectors, and $\|\cdot\|$ signifies the norm of vectors. Since the representation vector of the question title may lead to inaccuracies in the initially calculated question similarity, the accuracy is not optimal when compared to the actual statistical value. To align the calculated $\widehat{qs}_{ij}$ with the genuine statistical similarity qs_{ij} , we employ the loss function Ls for optimization according to Eq. (7). The representation of the question title ensures that the representation vector contains the question's relation, thereby enhancing the computation of relations between questions.

$$Ls(\widehat{qs}_{ij} - qs_{ij}) = \begin{cases} 0.5(\widehat{qs}_{ij} - qs_{ij})^2, & \text{if } |\widehat{qs}_{ij} - qs_{ij}| < 1 \\ |\widehat{qs}_{ij} - qs_{ij}| - 0.5, & \text{otherwise} \end{cases} \quad (7)$$

Subsequently, the threshold ω is applied to binarize the similarity. If the cosine similarity exceeds the threshold ω , $W_{QUE} = 1$. Otherwise, $W_{QUE} = 0$, forming a

binary adjacency matrix. This matrix, denoted as W_{QUE} , effectively represents the presence or absence of edges in a graph.

$$W_{QUE} = H(\widehat{qs}_{ij} - \omega) \quad (8)$$

Additionally, to enhance the question representation based on both the question association and difficulty, the two aforementioned loss functions are combined to create a new loss function for joint parameter optimization, as depicted in Eq. (9). This approach aims to enhance the robustness of the training process.

$$L_{sd} = \lambda_1 Ls(\widehat{qs}_{ij} - qs_{ij}) + \lambda_2 Ld(\widehat{qd}_i - qd_i), \lambda_1, \lambda_2 \in [0,1] \quad (9)$$

3.6 Assessment of student knowledge states based on GCN

3.6.1 Assessment

In our conceptualization, latent dependencies might exist among distinct exercises, and we perceive that the current states of a student's knowledge is intricately linked with their entire learning interactions over time. To dynamically delve into students' evolving knowledge states, we treat students' interactions as a sequential task. Following the notation from (Mao et al., 2023), we employ the tuple $x^t = \{q_i^t, c_j^t, a^t\}$ to characterize the relation between knowledge concepts and questions. Employing the individual exercise-response representation as input for the Gated Recurrent Unit (GRU), we sequentially capture and model the student's knowledge states. It can resolve the vanishing gradient issue and more effectively represent long-term dependencies in time series data.

$$z_k^t = \sigma(x^t W_{xz} + h_k^{t-1} W_{hz} + b_z) \quad (10)$$

$$r_k^t = \sigma(x^t W_{xr} + h_k^{t-1} W_{hr} + b_r) \quad (11)$$

$$\widetilde{h}_k^t = \tanh(x^t W_{xh} + (r_k^t \odot h_k^{t-1}) W_{hh} + b_h) \quad (12)$$

$$h_k^t = z_k^t \odot h_k^{t-1} + (1 - z_k^t) \odot \widetilde{h}_k^t \quad (13)$$

In the context provided, z_k^t , r_k^t , $\widetilde{h}_k^t$, and $h_k^t \in R^{N \times T}$ represent the vectors corresponding to the update gate, reset gate, candidate hidden states, and hidden states, respectively. Here, N denotes the dimensions of h_k^t . The weight parameters W_{xz} , W_{hz} , W_{xr} , W_{hr} , W_{xh} , and $W_{hh} \in R^{N \times T}$ are associated with the model, while b_z , b_r , b_h serve as bias terms. The Sigmoid activation function is denoted by σ , and the Hadamard product (element-wise multiplication) is represented by $\odot$. This statement succinctly describes the components involved in the GRU update mechanism to assess the hidden states of students.

3.6.2 Aggregate

The model aggregates the hidden states and embeddings for the answered concept i and its neighboring concepts $j \in N_i$. The aggregation method is shown in Eq. (14).

$$h_k^t = \begin{cases} [h_k^t, x^t E_x], & k = i \\ [h_k^t, E_v(k)], & k \neq i \end{cases} \quad (14)$$

Among them, $x^t \in \{0, 1\}^{2N}$ is an input vector of this layer, which represents the exercise answered correctly and incorrectly at time step t . 0 means the answer is wrong, and 1 means the answer is correct. $E_x \in R^{2N \times e}$ is a matrix embedding the knowledge concept index and response of answers, $E_v \in R^{2N \times e}$ is a matrix embedding the knowledge concept index, $E_v(k)$ represents the k -th row of E_v , and e is the embedding size.

3.6.3 Update

Subsequently, the model modifies the hidden states using the combined features and the structure of the knowledge states graph:

$$h_k^{t+1} = \begin{cases} f_{self}(h_k^t), & k = i \\ f_{neighbor}(h_i^t, h_k^t), & k \neq i \end{cases} \quad (15)$$

In the context above, f_{self} represents a multi-layer perceptron (MLP), while $f_{neighbor}$ denotes an arbitrary function dictating the dissemination of information to adjacent nodes according to the structure of a knowledge states graph. Although GKT effectively utilizes the inherent graph structure of knowledge for knowledge tracing purposes, this structure is often not explicitly provided. To operationalize the latent graph structure, $f_{neighbor}$ is defined according to Eq. (16). f_{out} and f_{in} represent MLPs.

$$f_{neighbor}(h_i^t, h_k^t) = A_{ik} f_{out}([h_i^t, h_k^t]) + A_{ki} f_{in}([h_i^t, h_k^t]) \quad (16)$$

3.7 Prediction

Lastly, the model produces the expected likelihood that a student will properly answer each knowledge concept at the subsequent time step. where σ is a sigmoid function, b_k is a bias term for node k , and W_{out} is a weight matrix shared by all nodes. The aim of the model's training is to reduce the observations' negative log-likelihood (NLL).

$$y_k^t = \sigma(W_{out} h_k^{t+1} + b_k) \quad (17)$$

The knowledge state from the nearby ideas can be gathered by using the edge information. Nevertheless, we verified that predicting y_k^t only on the basis of the

goal idea h_k^t knowledge state was preferable. Consequently, we restricted the application of graph structure information to the updating stage alone.

3.8 Model training

The goal of the KT apply is to forecast students' future responses with accuracy. Stated otherwise, the model must reduce the cross-entropy loss between the students' true label and the expected likelihood of properly answering the question. Thus, as demonstrated in Eq. (18), we use binary cross-entropy to minimize the loss of all predictive responses in order to optimize our model and train the learnable parameters.

$$L_{\text{cross-entropy}} = - \sum_t (r_k^t \log y_k^t + (1 - r_k^t) \log(1 - y_k^t)) \quad (18)$$

where r_k^t represents the true performance at time step t , and y_k^t represents the predictive performance at time step $t - 1$. The smaller the value of $L_{\text{cross-entropy}}$, the more accurate the results of predicting students' knowledge states assessments are.

4 Experimental results

In this section, we detail the experiments conducted on the ASSISTments dataset. We assess the efficacy of SGKT through evaluations using both predicted and actual responses from subsequent exercises. Additionally, we demonstrate the superior performance of SGKT by comparing it against several baseline models on the aforementioned dataset. Lastly, we present complementary analytical experiments to substantiate our innovative approach.

4.1 Datasets

To assess our model's performance, we conduct experiments using the ASSISTments 2009 "skill-builder" dataset,¹ a publicly available collection of students' mathematics practice logs offered by the online education service ASSISTments. This dataset is widely acknowledged as a benchmark in the fields of educational data science and knowledge tracing, frequently utilized to evaluate the performance of new models. It facilitates the comparison of new research findings with an extensive array of existing scholarly works. Additionally, the dataset provides annotations of predefined knowledge concepts for each exercise, offering detailed data that support in-depth analysis of students' mastery of knowledge concepts. We chose this dataset to ensure that our research results possess a high degree of comparability. Table 2 presents examples of existing knowledge concept tags within the dataset.

¹ [Online]. Available: <https://sites.google.com/site/assistmentsdata/home/2009-2010-assistment-data>.

Table 2 Example of knowledge concepts in ASSISTments

Knowledge concept ID	Knowledge concept name
2	Circle Graph
13	Median
48	Equivalent Fractions
70	Percent Of
...	...
193	Linear Equations
280	Addition and Subtraction Fractions
307	Volume Rectangular Prism

We apply specific conditions for preprocessing the dataset to optimize the experimental training effect. Initially, we filter out student user records with only a single answer or empty data in the "*skill_name*" column, aiming to eliminate noise. Subsequently, we aggregate simultaneous reply logs into sets to prevent artificially inflated prediction performance resulting from frequently co-occurring tags. Finally, we sort the dataset in descending order based on the number of answers for each student user in the "*user_id*" column and select the top 2,000 students' answer records. Through iterative experiments, we arrive at an optimal dataset. Following the application of these preprocessing conditions, we acquire 434,164 logs from the ASSISTments dataset, involving 2,000 students and 122 knowledge concepts.

4.2 Baselines and experimental settings

To evaluate the effectiveness and demonstrate the enhancements brought by the SGKT model, we implemented various state-of-the-art models for comparative analysis. The comparison models and their corresponding parameter settings are outlined below:

1. **DKT:** Piech et al. (2015) first proposed applying deep learning to KT and used RNN to model students' learning process. Our experiment followed the method of setting hyperparameters in DKT. The recurrent neural network's hidden layer size is set to 200, the batch size is configured as 32. Adam is chosen as the optimizer, and the learning rate is fixed at 0.001.
2. **DKVMN:** The memory network-based model aims to uncover the connections among knowledge concepts, providing direct assessments of a student's proficiency in each area. Hyperparameters are configured following the approach outlined by Zhang et al. (2017). The memory matrix has 20 columns, the hidden vector size is 32, with a batch size set to 32. Adam is chosen as the optimizer, and the learning rate is fixed at 0.001.
3. **GKT:** The model reframes KT as a time series node classification challenge within GNN, demonstrating outstanding performance and reliability. In comparison to previous KT models, GKT model introduces various methods for con-

structuring potential knowledge graph structures, including both statistics-based and learning-based approaches. Statistics-based approach includes Dense, Transition and DKT graph. Learning-based approach includes parametric adjacency matrix (PAM), multi-head attention (MHA) and variational autoencoder (VAE). These methods are better integrated and applied in the GKT model, resulting in significant performance advantages. We configured hyperparameters following the approach outlined by Nakagawa et al. (2019). All hidden vectors and embedding matrices are sized at 32, and the batch size is set to 16. Adam is chosen as the optimizer, and the learning rate is fixed at 0.01.

4. **GIKT:** The model employs GCN to consolidate problem embeddings and skill embeddings, addressing issues related to sparse data and multiple skills. We configured the hyperparameters following the approach proposed by Yang et al. (2020). The sizes of all hidden vectors and embedding matrices are set to 200 and 100, respectively, with a batch size of 32. Adam is chosen as the optimizer, and the learning rate is fixed at 0.001.

4.3 Ablation study

In this section, we perform an ablation study to test our model and determine the impact of each module. In order to perform experiments, we eliminate the relation between knowledge concepts and relevant question feature information in the ablation study. The details of the two settings are as follows.

1. **SGKT-NK:** The function of the relation between knowledge concepts in SKM is not loaded. Since we are unable to obtain meaningful relation labels between knowledge concepts, we conduct experiments using question embedding from the ASSISTments dataset.
2. **SGKT-NQ:** The question feature's functions are not loaded. We just keep the relations between knowledge concepts established based on SKM to illustrate the efficacy of the knowledge concept embedding we acquired, without taking into account the difficulty of each question or the relations between them.

4.4 Results and discussion

4.4.1 Performance analysis

We divided the dataset into training, testing, and validation sets in an 8:1:1 ratio. Through multiple experiments and using the area under the curve (AUC) and accuracy (ACC) as the final performance evaluation metrics, we evaluated the performance of each model, comparing the baseline with our model. Generally, higher AUC or ACC values indicate better predictive performance. According to the experimental results in Table 3, SGKT model showed a significant improvement in AUC and ACC metrics compared to other models, reaching 0.7794 and 0.7463, respectively. In comparison to the baseline model, the average

Table 3 Performance comparison of all KT methods

KT model	AUC	ACC
DKT	0.7276	0.7035
DKVMN	0.7200	0.7101
GKT(Dense graph)	0.7502	0.7329
GKT(Transition graph)	0.7435	0.7314
GKT(DKT graph)	0.7290	0.7268
GKT(PAM)	0.7513	0.7336
GKT(MHA)	0.7489	0.7351
GKT(VAE)	0.7474	0.7329
GIKT	0.7596	0.7454
SGKT-NK	0.7602	0.7360
SGKT-NQ	0.7619	0.7365
SGKT	0.7794	0.7463

performance of DKVMN was slightly lower than DKT, possibly due to the loss of relation information in the construction state of each knowledge concept.

For KT models based on various graph structures, we achieved a qualitative improvement compared to DKT model based on deep learning. AUC improved by approximately 5%, and ACC improved by nearly 4%. It is noticeable that, when not considering knowledge concept relations or question features separately, the average AUC decreased by nearly 2%, and the average ACC decreased by nearly 1%. This suggests that information about knowledge concept relations and question features positively influences the assessment of students' mastery of knowledge concepts. Tasks related to KT based on graphs are more focused on considering the relations and hierarchical structure between knowledge concepts. Therefore, adding information about knowledge concept relations may have a greater impact and lead to better results compared to adding a module for question features in answering.

Observing the loss values is crucial for evaluating model performance as it provides an intuitive reflection of how well the model fits the validation data. Lower validation loss generally indicates a better understanding of input data patterns and more accurate predictions. By comparing the loss trends on the ASSISTments validation set, we observed that SGKT model outperforms others with the smallest validation loss, as depicted in Fig. 7. Specifically, SGKT model excels by constructing SKM, modeling relations between knowledge concepts as a graph structure, and leveraging GCN and similar techniques to better capture question features, knowledge concept representations, and relations. This comprehensive modeling approach provides richer information to the model, optimizing predictive performance. In contrast, DKT model exhibits the largest validation loss, potentially due to limitations in modeling long-term student memory. Therefore, comparing the loss variations among different models helps gain insights into their relative performance, offering valuable information for further improvements and optimizations.

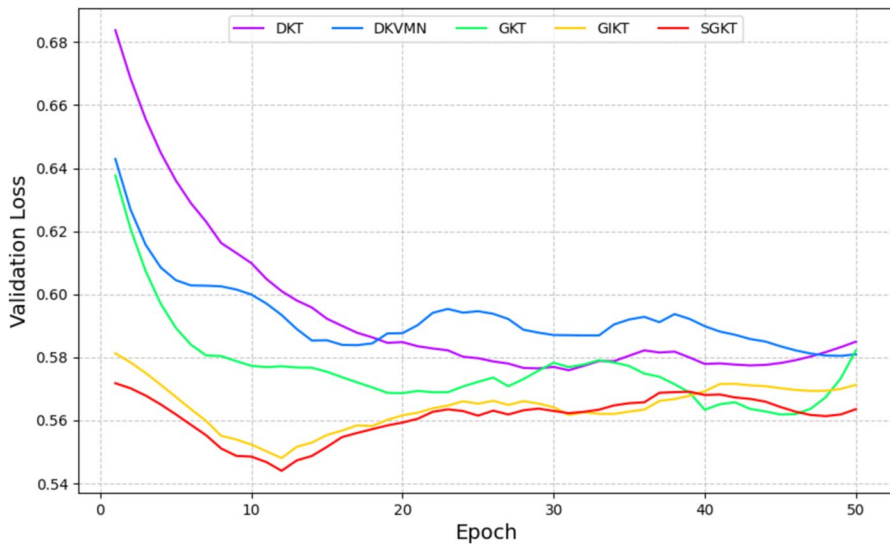


Fig. 7 Loss changes of all compared models on the ASSISTments validation set

Table 4 Question number and its corresponding knowledge concept

Skill_id	Skill_name	Problem_id	Sequence_id
10	Table	112,930	7891
16	Counting Methods	113,310	7892
17	Probability of Two Distinct Events	113,310, 113,258, 113,246	7892
18	Probability of a Single Event	113,039, 112,930, 113,029	7891
58	Addition Whole Numbers	53,673	5959
77	Finding Percents	109,014, 108,897, 108,948, 109,002, 108,957	6009
85	Absolute Value	53,673, 120,196, 120,164, 120,156, 120,232, 120,204, 120,172	8656
279	Multiplication and Division Integers	54,526	5967

4.4.2 Visualizing interpretability analysis

To conduct an in-depth investigation into students' mastery of knowledge, we conducted multiple experiments and ultimately selected 22 response records from a student (*user_id*=81441). These records encompass 8 crucial knowledge concepts {10, 16, 17, 18, 58, 77, 85, 279}, as detailed in Table 4. Each exercise may involve multiple knowledge concepts, and conversely, a knowledge concept may appear in multiple questions, all belonging to different question sets. Our research aims to unveil students' learning trajectories on specific knowledge concepts and their responses to new questions.

Fig. 8 Cosine similarity of knowledge concept relation

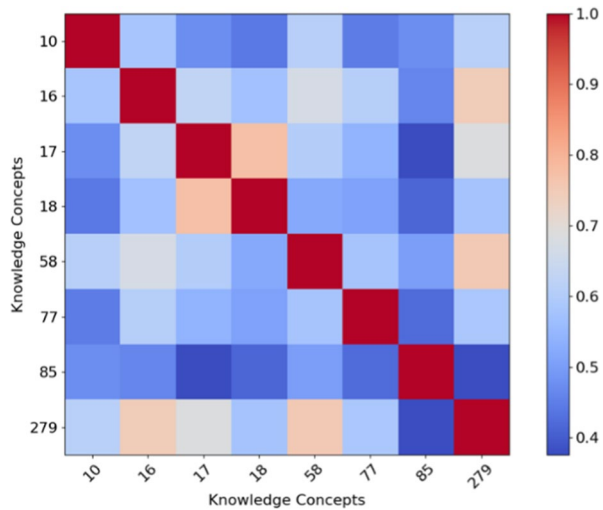


Fig. 9 Relational similarity of knowledge concept embedding vectors in SGKT model

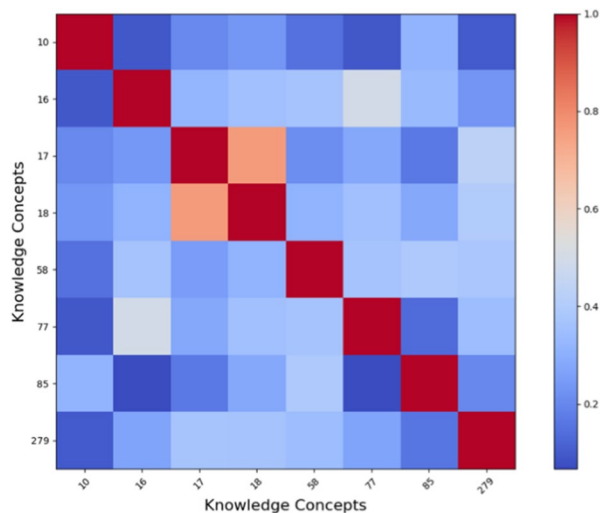


Figure 8 illustrates the cosine similarity between the embedding vectors of the 8 knowledge concepts, computed before model training. The numerical values within the squares represent the similarity between corresponding questions. In SGKT model, the embedding vectors implicitly contain rich information about the involved knowledge concepts, relations between knowledge concepts, etc. This facilitates more effective feature extraction by GCN. To validate this conclusion, Fig. 9 presents the calculated relations between embedded knowledge concepts. The observed increase in similarity between knowledge concepts 16 and 58 indicates that our constructed SKM more thoroughly considers relations between knowledge concepts. These two knowledge concepts both belong to the

higher-level knowledge category of "*Integers*", further emphasizing the effectiveness of the model in extracting relations between knowledge concepts and enhancing the interpret-ability of the model, thus improving its overall effectiveness.

Capturing a portion of the student's historical learning interactions, we employ SGKT model to track their real-time understanding of various knowledge concepts. The horizontal axis in the graph represents historical answer records, where each tuple (t, k^t, a^t) corresponds to the moment t , knowledge concept k^t , and the corresponding answer result a^t . The vertical axis displays the knowledge concepts tracked by the model. In Fig. 10, we use different colors to illustrate the student's mastery changes in these four knowledge concepts at different moments. This real-time tracing aids in promptly understanding the student's learning states and provides support for teaching.

Based on the experimental results, at the 4th and 13th time points, the student correctly answered questions related to knowledge concept 58, leading to an increase in the mastery level of that knowledge concept in the subsequent time intervals. Conversely, during the 8th, 10th, and 11th time points, when the student provided incorrect answers to questions involving knowledge concept \$10\$, there was a decrease in the mastery level of that knowledge concept in the following time intervals. This underscores the capability of SGKT to effectively capture the correlation between students' answering patterns and corresponding changes in knowledge proficiency.

The numbers in the graph ranging from 0 to 1 represent the model's predictions for the probability of a student correctly answering relevant questions in the next instance. To avoid confusion, only representative examples are displayed. From the 12th to the 20th time points, even without interacting with questions related to knowledge concept \$10\$, the corresponding knowledge proficiency levels are not constant. In the students' learning interactions, due to the interconnectedness of different knowledge concepts, potential and related knowledge proficiency levels are updated, leading to fluctuations in knowledge mastery. At the 27th time point, despite the student providing an incorrect answer, the knowledge mastery level increases. This is attributed to the challenging nature of the question, where even an incorrect response contributes to a certain degree of improvement in the student's proficiency.

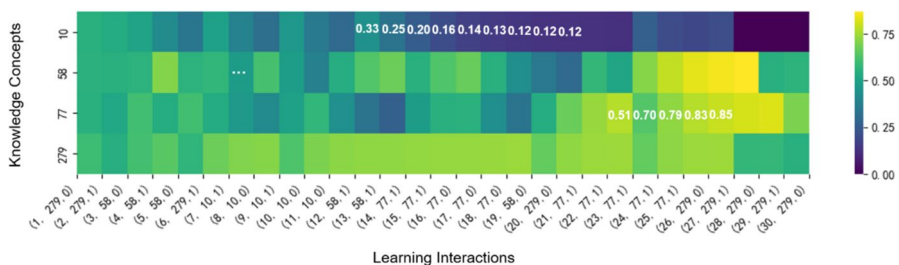


Fig. 10 Students visualize the prediction results based on SGKT model at different times

4.5 Future application prospects

4.5.1 Enhance learning assessment

The abilities and states of knowledge of students are implicit (Zhang et al., 2020). Real interactive data is produced by students while they solve questions. The act of tracing and monitoring learning can be made possible by uncovering hidden features and patterns in data. The challenge of data to evaluation conversion is thus successfully resolved. Prior to anything else, it is important to thoroughly examine students' internal cognitive mechanisms and to clarify each subject knowledge mapping and key concept assessment index system. Next, we extract learning features using KT correlation model we suggested. Finally, develop a learning dashboard applying technology (Schwendimann et al., 2017) to accomplish holographic learning portrayal and hasten the adoption of learning assessment methods in the field of education.

4.5.2 Boost learning services

The next generation of technology prompted expectations that it would provide new avenues for individualized instruction. The foundational component of the intelligent education platform is the knowledge tracing approach. The learning service assesses each student's unique cognitive differences and learning needs using improvement models related to KT. It then applies algorithmic models for matching, decision-making, and optimization. Finally, it offers students services like learning path planning (Chen, 2008), resource recommendations (Zhou et al., 2021), and academic states reports (Lin et al., 2020). Integrated system solutions, such online learning platforms and adaptive learning systems, offer pertinent features to aid students in understanding the reliability of scholarly analysis and the sense of recommended resources.

4.5.3 Direct teaching modifications

The personalization of education has been a desired goal in educational science and practice due to students' increasing dominance (Zhao et al., 2023). We suggested SKM that reconstructs the learning material and methodically arranges the knowledge context of each subject, creating new opportunities for teaching approaches based around knowledge. We developed an enhanced model connected to KT that can offer rich dynamic information. Based on this, accurate tracing can be used to identify the real issue, enabling teachers to dynamically modify and enhance the teaching method and related activities, supporting teachers in better guiding differentiated instruction and providing students with useful learning support scaffolds. It promotes the transition from "macro group" to "each single student" in individual teaching.

5 Conclusion

In this paper, we proposed the SGKT model, an innovative fusion of subject knowledge mapping that enriches the dynamics among knowledge concepts while also taking into account the complexity and interrelations of problems for an intricate scrutiny of student learning interactions. By weaving together knowledge concepts and question attributes within a graph-based knowledge tracing architecture, the SGKT model not only amplifies prediction accuracy and bolsters interpretability but also distinctly outperforms established methods, as evidenced through rigorous validation using the real-world ASSISTments dataset. This method not only yields deep insights into students' knowledge states, but also serves as an efficacious instrument for detecting learning hurdles and crafting tailored intervention strategies. The deployment of the SGKT model illuminates its vast potential within the educational technology sphere, heralding its positive implications for curriculum design and resource distribution. The ultimate goal is to provide every student with a personalized learning journey, thus enhancing learning efficacy and fostering educational fairness.

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Data availability The application as well, the data that were utilized and/or analyzed throughout the course of the current study can be obtained from the corresponding author upon making a reasonable request.

Declarations

Competing interests This manuscript has not been submitted to any other journal or other publishing venue, nor is it currently being reviewed by any other publication. Regarding the subject matter that is discussed in the manuscript, the authors do not have any connection with any organization that has a direct or indirect financial interest in the matter.

Conflict of interests The author declares no conflict of interest.

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